

ORIGIN AI: A Tiered Multi-Subject Context Engine for Low-Cost Pedagogical Mentorship

S Naveen (230101099)

Indian Institute of Information Technology Senapati, Manipur
Supervised by: Dr. Nongmeikapam Kishorjit Singh

Final Project Presentation
30 April 2026



Motivation: The Crisis of AI Scaling in EdTech

- **Financial Unsustainability:** High LLM API costs (up to Rs. 22,400/year per student).
- **Contextual Blindness:** Generic chatbots don't "see" what the student is looking at.
- **Pedagogical Drift:** Models often give direct answers instead of guided hints.

The Goal

Create an AI mentor that is **context-aware**, **pedagogically sound**, and **economically democratic**.

The Core Innovation: 4-Layer Context & Tiered Logic

4-Layer Context Assembly

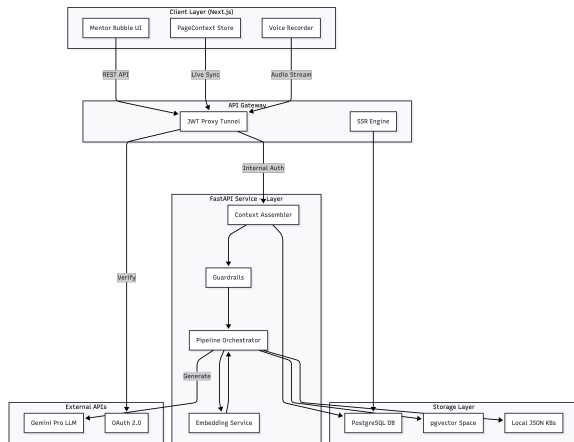
Fuses **Global** (History), **Page** (UI State), **Highlight** (Visual Selection), and **Query** into a single reasoning object.

Tiered Orchestration Pipeline

- 1 **Step 1:** Stored Solution Lookup (Exact Match)
- 2 **Step 2:** Subject Knowledge Base (Pedagogical Atoms)
- 3 **Step 3:** BFS Semantic Memory (pgvector Retrieval)
- 4 **Step 4:** LLM Generation (Fallback Only)

System Architecture Overview

- **Frontend:** Next.js (React)
- **Backend:** Python FastAPI
- **Persistence:** PostgreSQL + pgvector
- **Guardrails:** Scoring-based gating



Performance Benchmarks

Metric	Generic LLM	Origin AI	Improvement
Cost per 1k Queries	Rs. 1,245	Rs. 185	85% Reduction
Median Latency	2,400 ms	746 ms	73% Faster
Subject Accuracy	88%	96%	+8% Accuracy

- **Zero-Cost Hit Rate:** 60% of queries resolved without calling an LLM.
- **Pedagogical Score:** 4.8/5 vs 2.1/5 for generic models.

Societal Impact: Democratizing Elite Mentorship

- **Bridging the Digital Divide:** Brings premium 1-on-1 tutoring to remote learners at 1/10th the cost.
- **Student Engagement:** Transforms monotonous preparation into an interactive, "Tech-Native" experience (gamification).
- **Accessibility:** Voice-assisted instructions in English and Hindi for inclusive learning.

Innovation over Existing Tech

Moves beyond "Context-Blind" chatbots to an **In-Situ Mentor** that understands the student's live environment and historical mastery.

Key Takeaways

- Validated 85% cost reduction.
- 96% subject accuracy.
- Pedagogical guardrails enforced.

Future Areas to Focus

- **SLM Fine-tuning:** Moving to local Small Language Models (Gemma/Phi-2).
- **Offline Mode:** Compressed KBs for low-connectivity zones.
- **Visual Vision:** Point-and-ask for handwritten derivations.

Thank You!

Questions?